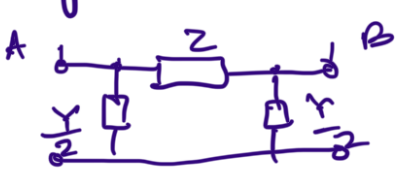
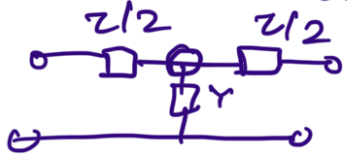
above 11kV



Syn. Gen. Transformer Transmission line



π- model



T- model

← leads to creation of one additional node

medium transmission line

long transmission line ← approximate π or T

Load ⇒ R, L, C usually Combination

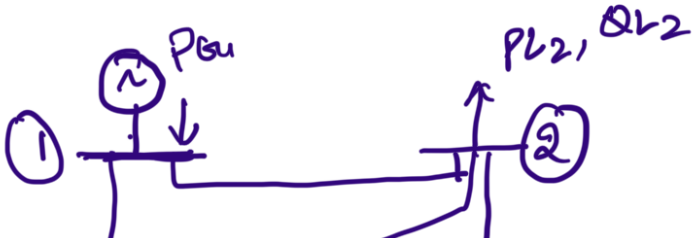


Solar PV

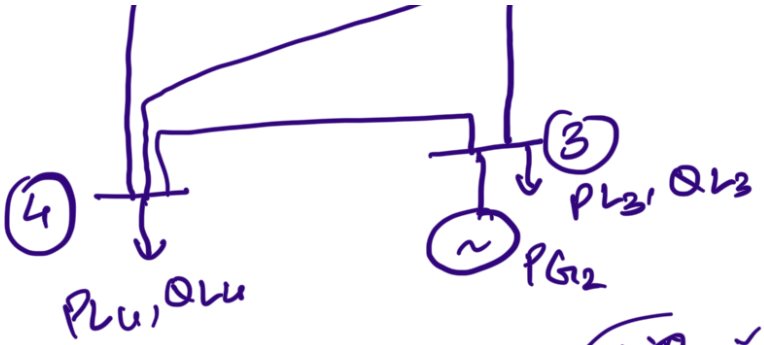


Induction Motor → load

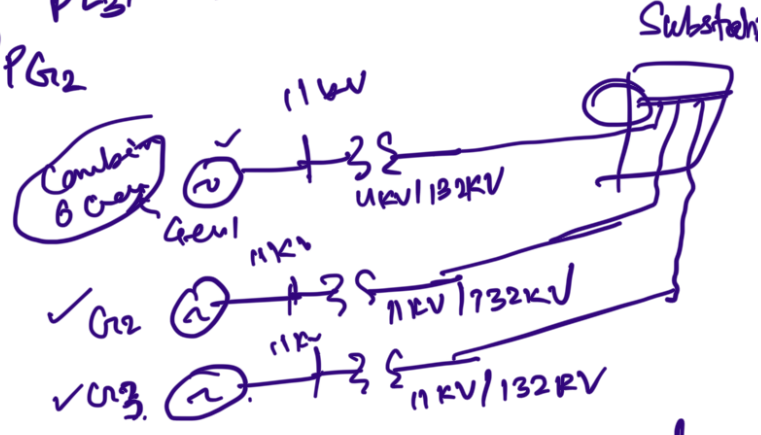




system at



132KV (...) or above



Gen 1 ⇒ Bus
↓
Giving Power into the network

Category of Buses ⟹ P_{Gen}, Q_{Gen} / P_{load}, Q_{load}, $|V|, \theta \rightarrow$ (5)

1,3 — unknown — unknown

Gen. Bus →
Load Bus → 2,4

If at a bus Power is injected in ← Gen Bus
- do — - taken out ← Load Bus

syc. op ≠ schedule

$P_i = P_{geni} - P_{loadi}$
$Q_i = Q_{geni} - Q_{loadi}$

$P_i = P_{geni} - P_{loadi}$

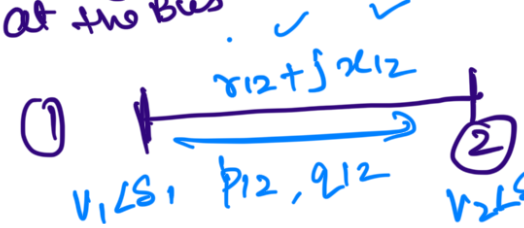
Bus Data

	P_i	θ_i	$\|V_i\|$	S_i
Gen Bus	known	unknown	known	unknown
Load Bus	known	known	unknown	unknown
Ref. Bus	unknown	unknown	known	$S_1 = 0$

2 unknown
2 unknown



$V_i \angle S_i$ ⇒ voltage at the Bus



← Analysis on the bases of gen and load

$S_{12} = P_{12} + jQ_{12}$

forecast for tomorrow (Day-Ahead)

Gen - forecasted

Load - forecasted

↳ system operator

Balancing and dispatch

Communicate back

System operator should ensure that power system operation is safe

What to be checked ?

→ voltages at all the Buses are within permissible limits

→ power flows in all lines and transformer etc. with permissible limit

Liberal Ut.

$$\begin{cases} V_{i\min} \le V_i \le V_{i\max} \Rightarrow 0.9 \le V_i \le 1.1 \text{ p.u.} \\ \Rightarrow 0.95 \le V_i \le 1.05 \\ S_{ij\min} \le S_{ij} \le S_{ij\max} \end{cases}$$

$\le S_{ij}$

$100 \text{ MVA} \le S_{12} \le 500 \text{ MVA} \Rightarrow$

Base 100 MVA

$1 \text{ p.u.} \le S_{12} \le 5 \text{ p.u.}$

$n \Rightarrow$ no of Buses
current example 4

$P_i, Q_i, |V_i|, \delta_i$

$2n$ — $2n$ variables are known
$2n$ variables are not known ,

→ an equation representing this network
↓
Solve them

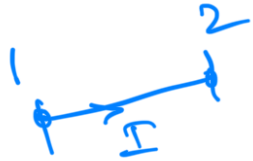
Bus				
1	$\|V_1\|$	δ_1	P_1	Q_1
2	$\|V_2\|$	δ_2	P_2	Q_2
3	$\|V_3\|$	δ_3	P_3	Q_3
4	$\|V_4\|$	δ_4	P_3	Q_4

One of Gen Bus ⇒ reference ← Slack Bus
↳ (1) $\delta_1 = 0$
$|V_1|, \delta_1 = 0$ ← unknown ← P_1, Q_1

(2) Balancing | scheduling

$$\sum_{i=1}^{N_G} P_{Gi} = \sum_{i=1}^{n} P_{Li} + P_{loss}$$

P_{loss} ← I^2R



$P_1 \Rightarrow P_{1sch} + P_{loss}$

Safe power system operation

$|V_i| \angle \delta_i$ P_i, Q_i

P_i, Q_i

2n → known
2n → unknown

4 Bus ↙ Bus Data

$|V_1|, \delta_1 = 0$
$|V_2|, \delta_2$
$|V_3|, \delta_3$
$|V_4|, \delta_4$

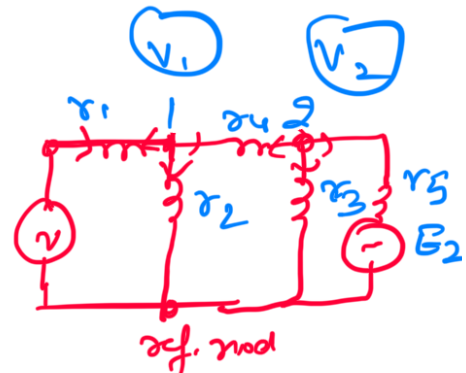
① Gen- load pattern / scenario ← load-gen. balancing
↓ sys. operator

② Topology → NIW

↳ Line data

	r_{ij}	x_{ij}	$(B_{ij}/x_{ij})/2$
1 – 2	r_{12}	x_{12}	$B_{12}/2$
2 – 3			
3 – 4			
4 – 1			
4 – 2			

$$I_p = \sum_{i,j \neq 1}^{n} Y_{ij} V_j$$



nodal equation at node ① E_1

nodal

$$\frac{V_1 - E_1}{r_1} + \frac{V_1}{r_2} + \frac{V_1 - V_2}{r_4} = 0$$

$$\frac{V_2 - V_1}{z_4} + \frac{V_2}{z_3} + \frac{V_2 - E_2}{z_5} = 0$$

Inspection

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Sum of admittances of all branches connected at bus 1

− admittance bet 1-2